

ACADEMIC GRANT

APPLICANT

Institution name

Università Campus Bio-Medico di Roma

Unit

Unit of Computer Systems and Bioinformatics

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CONTACT PERSON

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ACTIVITY

Name of the activity (max 200 characters)

AI in Health and Life Sciences Summer School

Description of the activity (max 1500 characters)

The "AI in Health and Life Sciences Summer School" at Università Campus Bio-Medico (UCBM) is a groundbreaking initiative designed to cultivate a generation of scholars adept at harnessing artificial intelligence to advance health and life sciences. Envisioned as a week-long intensive program, it will draw upon the respective strengths of UCBM and Umeå University to provide a comprehensive, interdisciplinary learning experience.

Goals of the Summer School:

- To equip participants with cutting-edge knowledge in AI applications in biomedicine, fostering innovation and research excellence.
- To create an interactive learning environment that encourages the exchange of ideas and collaborative problem-solving between Italian and Swedish students and faculty.
- To lay the groundwork for robust, long-term academic partnerships between Italy and Sweden, with a focus on AI's transformative potential in healthcare.

Planned actions:

- A curated lecture series delivered by esteemed faculty and industry professionals from both countries, covering AI fundamentals, bioinformatics, and precision medicine.
- Hands-on workshops in data analysis, machine learning, and AI ethics, complemented by case studies on health applications.
- A project incubator session, promoting collaborative proposals for joint research initiatives, aimed at solving pressing health challenges through AI.

Amount applied for, SEK (25,000 – 100,000 SEK)

100,000

Planned dates

September/October 2024

Number of participants

40

Field (choose one or more fields)

- ☐ Arts
- ☐ Humanities
- ☒ Natural Sciences
- ☐ Social Sciences
- ☒ Technology
- ☐ Other

Contribution to cultural exchange (max 1500 characters)

The Summer School is born as an Italy-Sweden academic partnership, offering a vibrant cultural exchange that transcends the conventional classroom experience. It is a melting pot where academic excellence meets cultural diversity, fostering mutual respect and understanding between the future leaders in AI from both nations. This School will be organized by Valerio Guarrasi (Assistant Professor at UCBM) and Paolo Soda (Professor at Umeå University), enhancing the Italy-Sweden tie.

Cultural Exchange Contributions:

- Joint seminars to explore the societal implications of AI in healthcare, considering the diverse ethical frameworks and health policies of Italy and Sweden.
- Cultural nights and social outings, curated to provide a taste of Italian and Swedish heritage, thereby enriching the participants' cultural awareness and building personal bonds.
- Formation of an Italy-Sweden AI in Health alumni group to maintain the exchange of ideas, cultural activities, and collaborative opportunities beyond the duration of the school.

This initiative is not just about developing AI expertise but also about weaving a fabric of intercultural dialogue, creating a foundation for enduring academic cooperation and a deeper appreciation of the unique cultural landscapes that shape how we approach technology in society.

Attachments

A one-page description of the activity detailing the topics covered, the actions/activities performed, and the projected outcomes for participants
(in English, PDF file)

Actions/activities

Description.pdf (1 pages)

A list of the budget items that the grant would cover, as well as the remaining budget covered by other funding (total amount is sufficient)
(in mEnglish, PDF file)

Budget items

Budget.pdf (1 pages)

PREVIOUSLY RECEIVED C.M. LERICI SCHOLARSHIPS OR GRANTS

Year(s)

Amount(s), SEK

1

23,183

CONFIRMATION OF INFORMATION

☒ I hereby declare that the information provided is true and correct.

GDPR CONSENT

☒ In compliance with EU General Data Protection Regulation (GDPR), I hereby authorize you to use and process my personal details contained in this application.

PhD Summer School in AI in Health and Life Sciences

Introduction

The PhD Summer School in AI in Health and Life Sciences is a collaborative educational initiative that will bring together students and professionals from Italy and Sweden to explore the innovative intersection of artificial intelligence and healthcare. This week-long program hosted by Università Campus Bio-Medico in Rome is designed, in cooperation with Umeå University, to stimulate an academic exchange that will enhance bilateral ties and lead to lasting collaborations in the field of AI.

Topics Covered

The curriculum is structured to provide a comprehensive understanding of AI in health sciences, covering:

- Fundamentals of AI and Machine Learning: Laying the groundwork with a strong theoretical foundation in AI.
- Data Science in Medicine: Approaches to managing and analyzing large datasets to improve patient outcomes.
- AI Applications in health and life sciences: methodologies and approaches in robotics, bioinformatics, natural language analysis, etc..
- AI Ethics and Policy: Discussing the ethical considerations and policy implications of AI in healthcare.
- Innovation and Entrepreneurship in AI: Exploring how AI can be utilized to create new ventures in the health sector.
- Soft skills: how to read and write a paper in AI research.

Activities

A blend of educational methods will be employed, including:

- Lectures and Seminars: Delivered by leading experts not only from UCBM and Umeå University, fostering knowledge exchange.
- Interactive Workshops: Practical sessions on AI tools, data handling, and analysis in clinical research.
- Panel Discussions: Cross-border expert panels to debate current trends and future directions of AI in health.
- Networking Events: Opportunities to meet peers and professionals, encouraging collaboration beyond the classroom.
- Cultural Activities: Organized events to explore the rich cultural heritage of Italy, enhancing the intercultural experience.

Projected Outcomes

The Summer School is structured to deliver a multitude of benefits:

- Academic Enhancement: Participants will gain advanced knowledge in AI applications in health, preparing them for cutting-edge research and professional careers.
- Research Collaboration: Facilitating the initiation of joint research projects between Italian and Swedish institutions.
- Professional Networking: Establishing a network of young professionals and experts to foster ongoing Italy-Sweden collaborations.
- Cultural Competency: Enhancing participants' understanding of different cultural and ethical perspectives in AI.
- Policy Impact: Providing insights that participants can take back to their home institutions to influence AI policy and practice in healthcare.

Conclusion

This Summer School represents a strategic opportunity to not only educate but also to build bridges between Italy and Sweden in the critical area of AI in health sciences. Through a rich blend of academic instruction, practical experience, and cultural exchange, participants will leave with enhanced capabilities, new partnerships, and a global perspective.

Budget

Introduction:

We are requesting a total of 100,000 SEK to cover critical costs associated with the successful execution of the Summer School. These costs are instrumental in facilitating the academic exchange and providing a rich learning environment for participants.

Table of Costs:

#	Item	Amount (SEK)
1	International flights for international speakers	30,000
2	Local transportation for field trips	5,000
3	Lodging for international speakers	20,000
4	Scholarships for 10 Swedish students	25,000
5	Lecture halls and workshop rooms rental	20,000
	Total Grant-Covered Costs	100,000

Justification of Costs:

Each cost element has been carefully considered to ensure the effective delivery of the Summer School.

Travel and accommodation expenses are essential for the participation of leading Swedish experts, which is a cornerstone of the program. Speaker fees are modest but necessary to attract high-caliber professionals to deliver the curriculum.

To ensure the Summer School's accessibility and to reinforce the Italy-Sweden academic connection, we have earmarked 25,000 SEK of the grant to provide full scholarships for 10 deserving Swedish students. These scholarships are a critical component of our cultural exchange initiative, designed to remove financial barriers and promote diversity within the cohort. The scholarships aim to attract talented students with a demonstrated interest and aptitude in AI applications within health and life sciences. A merit-based application process will be implemented, requiring students to submit their academic records, CV, and a personal statement outlining their interest in the program. A joint committee from UCBM and Umeå University will review applications to ensure a fair and unbiased selection process. The scholarship funds will be directly applied to cover the school fees for the selected students, simplifying the administrative process and ensuring transparency.

Other Funding Sources:

The Summer School could receive support from potential private sector sponsors, but we also plan to open applications to other international PhD students, who will pay a registration fee. These contributions will cover additional operational costs and ensure that the program is accessible to a wider range of participants. We will also apply for financial support from IEEE Technical Committee on Life Science.

Sustainability Plan:

If the success rate of the first edition of the Summer School is sufficient, this initiative will be planned as an annual event. Future editions will be supported by a combination of registration fees, continued institutional support, and active pursuit of external funding sources.